

Baby-Step Giant-Step Algorithm for DLog

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Objective

Find scalar k such that $Q = kP$ on an elliptic curve group of order n .

Baby-Step Giant-Step Algorithm

Key insight: Express k as $k = im + j$ where $m = \lceil \sqrt{n} \rceil$, and both i and j range from 0 to $m-1$.
Algorithm

- Rearrange the equation: $Q = kP = (im + j)P = i(mP) + jP$
- Rearranging: $Q - jP = i(mP)$
- Baby steps: Compute and store all values $Q - jP$ for $j = 0, 1, 2, \dots, m-1$
Store these in a hash table: $Q - jP \rightarrow j$
- Giant steps: For $i = 0, 1, 2, \dots, m-1$:
 - Compute $i(mP)$
 - Check if this value exists in your baby-steps table
 - If yes, you found a match: $i(mP) = Q - jP$
 - Therefore: $Q = i(mP) + jP = (im + j)P$
 - So $k = im + j$
- Complexities:
 - Time: $O(\sqrt{n})$ - we do at most $\sqrt{n}$ baby steps + $\sqrt{n}$ giant steps
 - Space: $O(\sqrt{n})$ - we store at most $\sqrt{n}$ baby-step results

Why $m = \lceil \sqrt{n} \rceil$ is optimal: It balances computation vs storage. If m is smaller, you need more giant steps; if larger, you need more baby steps and more storage.